The Methods of Improving the Manufacturing Resource Planning(MRP Ⅱ) in ERP

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***Abstract*-This paper focus on the methods to improve the traditional MRP Ⅱ process according to the requirement of the time base competition. By finding out the defections in the process of traditional MRP Ⅱ, such as using the fixed lead-time, using the fixed secure stock, and lacking of time management, then we building 5 models and integrating these models into the process of MRP Ⅱ to make all the parameters which correlate with the environment of the enterprise become dynamic parameters, and make the MRP Ⅱ obtain the ability of time management.**

Key words: improve, MRP Ⅱ, method, model, TBC

Ⅰ INTRODUCTION

Since 1990s, the market has changed greatly. As the competition become severer, the life cycle of products become shorter, and the demand of customers become more multiples, the competitive advantage of manufacturing have to be shifted form price, quality, and service to the adaptability of manufacture. The one who have grasped time in the market would predominate in the competition. All of these mean that we are now in the "Time Base Competitive age". In such a competitive environment, how to reduce the time of manufacturing, and put the products which loved by customers into the market has became a very important advantage in competition.

The MRP Ⅱ (Manufacturing Resource Planning) which born in America at the end of 1970s is a very good method which has been used in enterprise resource and manufacturing management. Now, the MRP Ⅱ is the core of ERP(Enterprise Resource Planning) System which has been widely used in manufacturing industry. Nevertheless, there are some defections in the method of MRP Ⅱ used to form manufacturing plans in the Time Base Competitive environment, such as the setting of fixed lead-time, the setting of fixed secure stock, lacking of guidance in capability resource planning, and ignores the time management in the manufacturing management.

This paper focus on improving the process which MRP Ⅱ used to form manufacturing resources plans, and resolving the problems exist in the MRP Ⅱ at the TBC age. The final purpose of this paper is to improve the MRP Ⅱ by integrating 5 models into the process of forming plans, and make the thoughts of TBC combined with the theory of MRP Ⅱ.

Ⅱ THE DEFECTS OF TRADITIONAL MRP Ⅱ IN TBC

As the competitive environment of market become more and more uncertain, the competition of enterprises shifted to time base competition, the time become the most powerful resource which make the enterprises to gain advantage in competition. In the Time Base Competitive environment, how to produce the goods wanted by customers in the shortest time with the lowest cost and how to guarantee sufficient supply in the lowest level of stocks become the most important points in manufacturing management. So the defections of traditional MRP Ⅱ which born at the end of 1970s become more and more obvious, the defects are listed underneath:

1) **The traditional MRP Ⅱ make the system administrator having the tendency to set a longer lead-time which used to form manufacturing plans.**

The lead-time is a very important parameter in the MRP Ⅱ

The topic of this paper from the China National Social Science Fund project, the approval number of this project is: 07AJY024, and this project is the "211" key project engineering (211-6-44) of China Shanghai University of Finance and Economics,

978-0-7695-3521-0/09 $25.00 © 2009 IEEE
DOI 10.1109/ICCET.2009.18

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system. Usually, the administrators like to set a longer lead-time, because using the longer lead-time make the MRP Ⅱ system form plans whatever how big or small the demand is. The longer lead-time doesn't reflect the real capability of manufacturing, and the longer lead-time increase the difficulty of market forecast that would increase the risk of manufacturing.

The using of longer lead-time to form plans would also increase the unfinished products' stocks and secure stocks, these stocks occupy too much circulating capital what would decelerate the running speed of logistic system. Even worse, the longer lead-time covers up the resource restriction existing in the process of manufacturing, which go against the wish of the enterprise to build an adaptability manufacturing system.

2) The traditional MRPⅡ use the fixed lead-time as parameters, that do not satisfy the requirement of building a adaptability manufacturing system.

The lead-time was set by the administrators when initialized the MRPⅡ system, and the lead-time in the MRPⅡ system would not be changed when the initialization had finished. So the traditional MRPⅡ ignores the situation that the using of elastic workers and the advance of manufacturing capability would reduce the time of manufacture when using the fixed lead-time to form manufacturing plans.

3) The precondition of forming the manufacturing plans is the assumption of infinite manufacturing capability.

The traditional MRPⅡ system ignores the restriction of manufacturing capability when forming the plans. So the CRP (Capability resource planning) is a capable adjustment after event, this adjustment is a complex and difficult task, and it lack of guidance. The adjustment of capability without guidance would make the administrators harder to decide how to distribute the finite resource, and it often makes the administrators adjust the capability without considering time reduction.

4) The traditional MRPⅡ lack of restriction on secure stocks.

The secure stocks are the stocks set in order to prevent the halt of manufacturing system which cause by the fluctuation of demands, and the secure stocks correlate with the fluctuant degree of demands. The quantity of secure stocks is another parameter set when the MRPⅡ system is initialized. Like the lead-time, the quantity of secure stocks is also a fixed parameter in traditional MRPⅡ system. It ignores the situation that the demand is fluctuant. It should set bigger secure stocks when the fluctuant degree of demand was big, and set smaller secure stocks when the fluctuant degree of demand was small.

Ⅲ IMPROVE ASSUMPTION

In order to make the improvement more strictly, there are some assumptions of the enterprise environment that should be set first:

1) The running environment of MRPⅡ is MTO (Make to Order).
2) For the finished goods the MRPⅡ use the direct quantity; For the semi-finished goods or purchase material the MRPⅡ use the fixed quantity, and the fixed quantity is the most economical quantity.
3) For the finished goods and semi-finished goods the companies do not set the secure stocks.
4) For the purchase material the companies set the secure stocks to prevent the halt of manufacturing system.
5) The demands of goods satisfy the normal distribution in the planning period, and the demands are all independent.

Ⅳ IMPROVE METHOD AND THE MODELS

In order to satisfy the requirement of time base competition and improve the traditional MRPⅡ system, there are five models should be integrated into the traditional MRPⅡ system. These five models are:

A. The Model of Resource Balance in Main Production Scheduling

1） The description of the model

The demands of customers are changeable in the time base competitive environment, if the company keeps too many finished products, it would suffer great losses when the demands changed greatly. So the model of resource balance should be integrate into the MRPⅡ system when it forming the MPS(Main Production Scheduling) plan. The resource balance model is a model that balances the finite resource while fully considering reducing the inventory of finished products and lowering the production costs. The purpose of this model is to improve the MPS, and make the MPS satisfy the requirement of TBC.

2） The description of the symbol in the model

D_{ik} - The demand of the product i in the period k; P_{ijk} - The quantity of product i that produce in the period j in order to

satisfy the demands in the period k; P_{ij} -The quantity of i product that produce in the period of j; Wt_{ij} -The demand of normal working hours for producing the i product in the period of j; Ot_{ij} -The demand of overtime working hours for producing the i product in the period of j; Wt_j -The restriction of normal working hours in the period of j; Ot_j -The restriction of overtime working hours in the period of j; Wp -The unit cost of normal working hours; Op -The unit cost of overtime working hours; S_i -The unit inventory cost of the i product; C_i -The manufacturing cost of i product; n_i -The working hours for producing per unit i product; n -The variety of products; T -The number of the periods。

3） The model of resource balance

$$\min Z=\sum_{i=1}^{n}\sum_{j=1}^{T}\sum_{k=j}^{T} C_i P_{ijk}+\sum_{i=1}^{n}\sum_{j=1}^{T}\sum_{k=j}^{T}(k-j)S_i P_{ijk}+\sum_{i=1}^{n}\sum_{j=1}^{T} Wp\times Wt_{ij}+\sum_{i=1}^{n}\sum_{j=1}^{T} Op\times Ot_{ij}$$

$$\begin{cases} \sum_{j=1}^{k} P_{ijk}=D_{ik} \quad i=1,2,.....,n;k=1,2,.....T \\ \sum_{k=j}^{T} n_i P_{ijk}\le Wt_{ij}+Ot_{ij} \quad i=1,2,.....,n;j=1,2,.....,T \\ \sum_{i=1}^{n} Wt_{ij}\le Wt_j \qquad j=1,2,.....,T; \\ \sum_{i=1}^{n} Ot_{ij}\le Ot_j \qquad j=1,2,.....,T; \\ P_{ijk},Wt_{ij},Ot_{ij}\ge 0, P_{ijk} \text{ is integer} \end{cases}$$

The P_{ijk} , Wt_{ij} and Ot_{ij} is the decision variable in the model. The P_{ij}($P_{ij}=\sum_{k=j}^{T} P_{ijk}$) is the decision variable that we want get finally, it tell us how many i products should be produced in the j period.

B. The Model of Secure Stock Forecast

1) The description of the model

As we know the quantity of secure stock is fixed in the traditional MRP Ⅱ system, that is one of its defects. So we should integrate the secure stock forecasting model in to the traditional MRP Ⅱ system to make the secure stock become a dynamic parameter. The model is a statistical model that uses the fluctuation of demand to forecast the secure stocks.

2) The description of the symbol in the model

P_j -The average total demands of j purchase material within a certain period; R_{ij} -The average demand of i product which correlate with the j purchase material within a certain period; σ_{ij} -The standard deviation of demand of i product which correlate with the j purchase material within a certain period； Ω_j -The standard deviation of total demands of j purchase material within a certain period； n_{ij} -The ratio of demand between the i product and the j-related material; S_j -The number collection of the product which relate to the j purchase material; CSL -The probability that there is no stagnation in the production within the production cycle； $\overline{LT_j}$ -The time interval of procuring the j material; LTs_j -The shortest time interval of procuring the j material； $\overline{LTm_j}$ -The average time interval of procuring the j material； LTl_j -The longest time interval of procuring the j material; ss_j -the secure stock of j material； B-The collection of all procuring material

3) The model of secure stock forecast

This model is base the assumption that the distributions of the demands are the normal distribution and all these distribution are independent.

$$ss_j=\overline{LT}\times P_j+\sqrt{\overline{LT}}\times\Omega_j\times F^{-1}(CSL) \qquad j\in B;$$

$$P_j=\sum_{i\in S_j} n_{ij}R_{ij};$$

$$\Omega_j=\sqrt{\sum_{i\in S_j} n_{ij}\sigma_{ij}^2};$$

$$\overline{LT_j} = \frac{LTs_j + 4\overline{LTm}_j + LTl_j}{6};$$

C. The Model of Producing Lead-time Calculates

1) The description of the model

The producing lead-time is a fixed parameter in the traditional MRP Ⅱ system. The model of producing lead-time calculates is the model use to calculate the producing time according to the capability of manufacturing system. So if we integrate the model into the MRP Ⅱ system, it would make the producing lead-time become a dynamic parameter which changes when the environment of manufacturing changes.

2) The description of the symbol

i -The No. of working center; P -The collection of all products and purchase material; S_j -The No. collection of the working center that the j material involved; Pt_i - The preparing hours for the i working center prepare to process per product or material; Mt_i - The working hours for the i working center to process per product or material; Qt_i -The waiting time before the i working center process the material; Tt_i -The time for the i working center to transport the material or product to the next working center; n_k -The production volume of the k product; m_{kj} -The ratio of requirement between the k product and the j material; R_j -The collection of the product which need the j material; W_i -The rated working hours of the i working center per day; LTp_j -The production lead-time of j product or material, the unit of this is the day; PLT_j -The production lead-time of j product or material, the unit of this is the week; WT -The per week working hours in the working calendar.

3) The model of producing lead-time calculates

$$LTp_j = \sum_{i \in S_j} (Qt_i + \left[\frac{\sum_{k \in R_j} n_k \times m_{kj} \times (Pt_i + Mt_i)}{W_i} \right] + Tt_i) \quad j \in P;$$

The symbol-[ ] means give the smallest interger greater than or equal to the result of formula

$$PLT_j = \left[\frac{LTp_j}{WT} \right] \quad j \in P$$

The symbol-[ ] means give the smallest interger greater than or equal to the result of formula

D. The Model of Purchase Lead-time Forecast

1) The description of the model

The purchase lead-time is another fixed parameter in the traditional MRP Ⅱ system, but the purchase lead-time would be change when the condition of purchase is changed. The model of purchase lead-time forecast is the model that using the BP neural network to forecast the purchase lead-time according to the real procuring condition. The first step of building this model is to build a BP net work, then using the training set to train the network. When the neural network had been trained, we can use the network to forecast the lead-time of purchase.

2) The step to build the model

The Fist step: Using the method such as fishbone figure or other analytical tools to find out the factors that affect the purchase in the real procuring environment. Then quantify these factors, and let them to reflect the real purchase condition.

The Second step: According to the effect factors to construct frame of the training set of different purchase material. Then build the training sets according to the history of purchase. These training sets are used to train the BP neural networks.

The Third step: Build the structure of the BP neural network. According to the demand of the forecast, choosing the structure that contain tree hidden layer, multi-input and single output (see the Figure 1)

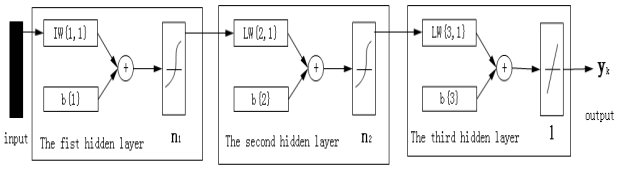


Figure 1 the structure of the BP neural networks

The Forth step: Determining the transmission function in the hidden layer of the BP neural networks.

In the first and second hidden layer, we use the tansig function as transmission function.

$$a(n) = \tan sig(n) = \frac{1-e^{-n}}{1+e^{-n}}$$

In the third hidden layer, we use the purlin function as transmission function.

$$a(n) = purelin(n) = n$$

The Firth step: Using the train set to train the neural network what we have built. After training, we can use the network to forecast the purchase lead-time, and all we have to do is just input the indexes which represent the purchase condition.

3) Example

Table1. The train set of the model

No.	X1	X2	X3	X4	X5	X6	Lead-time
1	100	0. 05	50	air	1	1500	5
2	200	0. 10	150	air	1	1250	5
3	100	0. 03	30	train	2	1750	8
4	200	0. 05	80	train	1	1250	6
5	100	0. 10	100	air	1	700	2
6	100	0. 05	50	train	1	1500	5
7	200	0. 04	75	air	1	1500	4
8	300	0. 05	100	ship	2	1500	11
9	100	0. 06	50	Ship-truck	1	1500	9
10	100	0. 05	50	train	1	1500	5
11	120	0. 08	75	Ship-truck	2	450	6
12	240	0. 1	120	train	4	450	5
13	150	0. 07	90	air	2	500	4
14	120	0. 10	120	truck	1	500	3
15	360	0. 10	150	truck	2	500	6

Table 1 is a part of the train set for the model of one material's purchase leading-time forecast. All of the data we in the table we gain in a company. And according to the investigation, there are six factors would affect the purchase leading-time of the company for one of is purchase material. X1: the purchase quantity of the material; X2: The condition of the work force about the company which produce the material; X3: the efficiency of the work force about the company which produce the material; X4: The transportation method. X5: the weather condition. X6: distance. And the table 1 is just a part of the training set.

We have the training set already now, then we can use this train set to train our neural network. We use the Matlab to build this model.

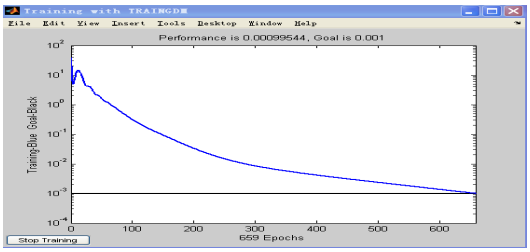


Figure 2 the curve of training performance

Now we have the neural network which had been trained, then we can use the model to forecast the purchase lead-time of the material.

e.g we put the index X_1=1.0;X_2=0.05;X_3=0.5;X_4=4 X_5=1;X_6=0.5 in to the model, we will get the result that the purchase lead-time of this material in this purchase condition is 3.3 days.

E. The Model of Resource Balance in Material Requirement Plans

1) The description of the model

The model of resource balance in the MRP is a model that integrates the technology of PERT and CPM into the MRP. The model considers the manufacturing activity as a project, then using the BOM list and the lead-time to form a net of the manufacturing plan. By using the network we can find out the critical path, and then we can balance the resource of manufacturing. In addition, we can implement the time management in manufacturing system.

2) The operation step of the model

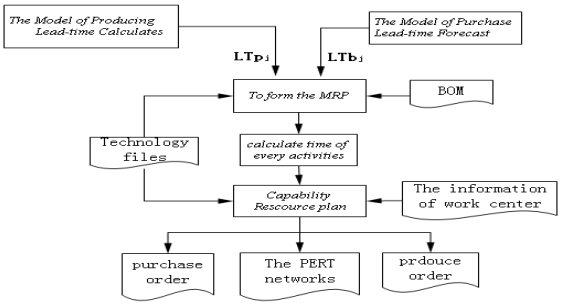


Figure 3 the operation step of the Resource Balance model

The First step: Considering the activity of manufacturing as a project and all the results of produce and purchase material as nodes. Then form the networking of manufacturing plan by using the results of MRP and the results of lead-time forecast. The line between two nodes

represents the producing activity or the procuring activity.

The Second step: Using the result of producing lead-time calculating and the result procuring lead-time forecasting as the time need of activities ($t(i,j)$) in the networks.

The Third step: Set $t_E(1)=0$, then using the formula to calculation:

The earliest time to receive the material or product j

$$t_E(j)=\max_{(i,j)\in I}\{t_E(i)+t(i,j)\}$$

The latest time to receive the material or product i

$$t_L(i)=\min_{(i,j)\in I}\{t_L(j)-t(i,j)\}$$

The earliest start time of the activity (i, j)

$$t_{ES}(i,j)=t_E(i)$$

The earliest finish time of activity (i, j)

$$t_{EF}(i,j)=t_E(i)+t(i,j)$$

The latest start time of the activity (i, j)

$$t_{LS}(i,j)=t_L(j)-t(i,j)$$

The latest finish time of activity of (i, j)

$$t_{LF}(i,j)=t_L(j)$$

The Forth step: Finding out the critical activity

$$S(i,j)=0, S(i,j)=t_{LS}(i,j)-t_{ES}(i,j)$$

Then calculate the time need of entire project, and calculate the relaxation time of the non-critical activities:

$$r(i,j)=t_E(j)-t_E(i)-t(i,j)$$

The Fifth step: Calculating the starting time of the entire project according to the time of the critical activity. Then consider balancing the manufacturing resource without prolong the time of the critical path at first. If the resource can not be balanced in the time of the critical path, then considering prolong the time of the critical path. Then amend the networks and the MRP plans.

The sixth step: Distributing the networks and the MRP plans which had been amended after the resource balance to the department of purchase and the department of manufacture.

Ⅴ INTEGRATE THE FIVE MODELS INTO THE MRP Ⅱ

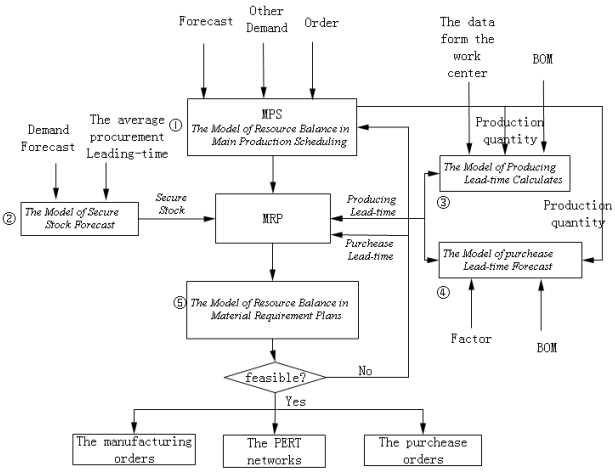


Figure 4 The operational model of the improved MRP Ⅱ

Now, we considering how to integrate these five models into the MPR Ⅱ, figure4 show the operational model of the improved MRP Ⅱ.

We integrate *the model of resource balance in main production scheduling* into the process of MPS. When the new MRP Ⅱ forming the MPS, it consider the resource restriction meanwhile. So the process of forming the MPS is the process to balance the resource in the planning period and calculate the most economical quantity of the finished products which produce on each week or month. Here, the economical quantity is the quantities which meet the demand in every week or month with balancing the resource restriction in every week or month, and it is the quantities which are set by fully considering reducing the inventory of finished products and the production costs.

Base on the MPS which have been formed, the MRP Ⅱ begin to form the MRP(material resource planning). We then integrate *the Model of Secure Stock Forecast, the Model of Producing Lead-time Calculates, and the Model of Purchase Lead-time Forecast* into the process of MRP Ⅱ to form the MRP.

The secure stock forecast model fetches the demand forecast information from the ERP, and calculates the average purchase leading-time by using the history purchase data, after having all the information the model need, it would work out the most appropriate secure stock of every purchase material. All of these secure stock quantities would be used in forming

the MRP. By integrate this model into the MRPⅡ make the secure stock which is a fixed parameter in traditional MRPⅡ become a dynamic parameter that correlate with fluctuation of demand.

The Model of Producing Lead-time Calculates fetches the information of the work center and the BOM from the file stored in the ERP system, and receives the producing requirement of one particular product in a period from the MPS. Then, the model can work out the producing lead-time of every producing activity. After we integrate this model into the MRPⅡ, while the productivity of the manufacturing system changing or the produce quantities changing, the producing leading-time will change. The leading-time become a dynamic parameter in the new MRPⅡ, this make the leading-time more accurate to reflect the time requirement for producing. This make the producing orders which formed by MRP have more value of guidance.

The purchase lead-time would be change when the condition of purchase is changed. So we integrate *the Model of Purchase Lead-time Forecast* into MRPⅡ, and before forming the purchase material order, the system first ask for the information of the purchase condition. The model fetches some part information from ERP, and call for information which can't be obtained in the ERP system form the purchase department, such as asking the employee of the purchase department the question like "which vehicle to choose to transport the purchased material?", "what the weather would like during the purchase period?" After having enough information this model need, it can forecast how many days this purchase order would take. So after integrating this model into the process of MRP, the purchase lead-time of one particular material become a dynamic parameter which changes while the condition of material purchase changes.

The last model we want to integrate into the MRPⅡ process is *the Model of Resource Balance in Material Requirement Plans*. We combine this model with the CRP (capability resource planning). Combine this model with the CRP, we now can find out the time requirement of every production activity, and find out when the production activity begin or finish, then we can form the PERT networks which contain the time information of each production activity and information about the relation among production activities which are contained in the process of producing one particular product. By using the PERT networks, the system can easily find out the critical activities and the shortest time needed to produce every product. Then, by using the technology of resource balance of PRET and CPM the system can balance the resource distribution just adjusting the relaxation time of the non-critical activities. These balance processes wouldn't increase the shortest time required to product the product. After resource balance, the system can delivery the PERT producing activities networks of every product every workshop. By doing this, the manager of each workshop can know exactly when the activities that belong to their workshop beginning, even when these activities should be finished. To form the producing activities networks and delivery those to the workshop make all workers in the workshop understand how important they are, because they can see the fact that the efficiency of their workshop determines the on time finishing of the whole manufacture project. And this reflects the concept of management by objectives.

Ⅵ Conculusion

The purpose of this paper is to provide methods to improve the traditional MRPⅡ system according to the demand of the time base competition. The manufacturing lead-time, the purchase lead-time and the secure stock become dynamic parameter in the MRPⅡ by integrating the 5 models into the traditional MRPⅡ, and by integrating the Model of Resource Balance in Material Requirement Plans into the MRPⅡ make the ERP obtain the ability of time management in manufacturing management.

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